

Total No. of Questions :6]

SEAT No. :

P66

OCT. -16/BE/Insem. - 120

[Total No. of Pages :1

B.E. (Mechanical)

MACHINE TOOL DESIGN

(2012 Course) (Semester - I) (Elective - I) (402044D)

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Neat diagrams must be drawn wherever necessary.
- 2) Assume suitable data, if necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of non-programmable electronic calculators is allowed.

Q1) Explain structural diagram of speed box with broken geometrical progression. [10]

OR

Q2) Design the feed box of the Lathe machine for the feed range 0.05 - 4 mm/rev. Given $\phi = 1.41$. Assume a suitable kinematic train between the spindle and the input shaft of the feed box, and between the output shaft of the feed box and the rack, attached to the underside of the lathe bed. [10]

Q3) Explain design procedure of Column with respect to coefficient of distribution of shearing displacement. [10]

OR

Q4) Discuss the factors affecting stiffness of machine tool structure and the methods of improving it. [10]

Q5) Explain the design procedure of Hydrodynamic slide-ways. [10]

OR

Q6) From among the slideways combination flat-flat, V-flat, V-V of a Lathe, which one would provide for the least radial deflection? Give your conclusion with mathematical proof. [10]

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